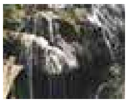
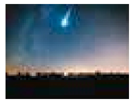
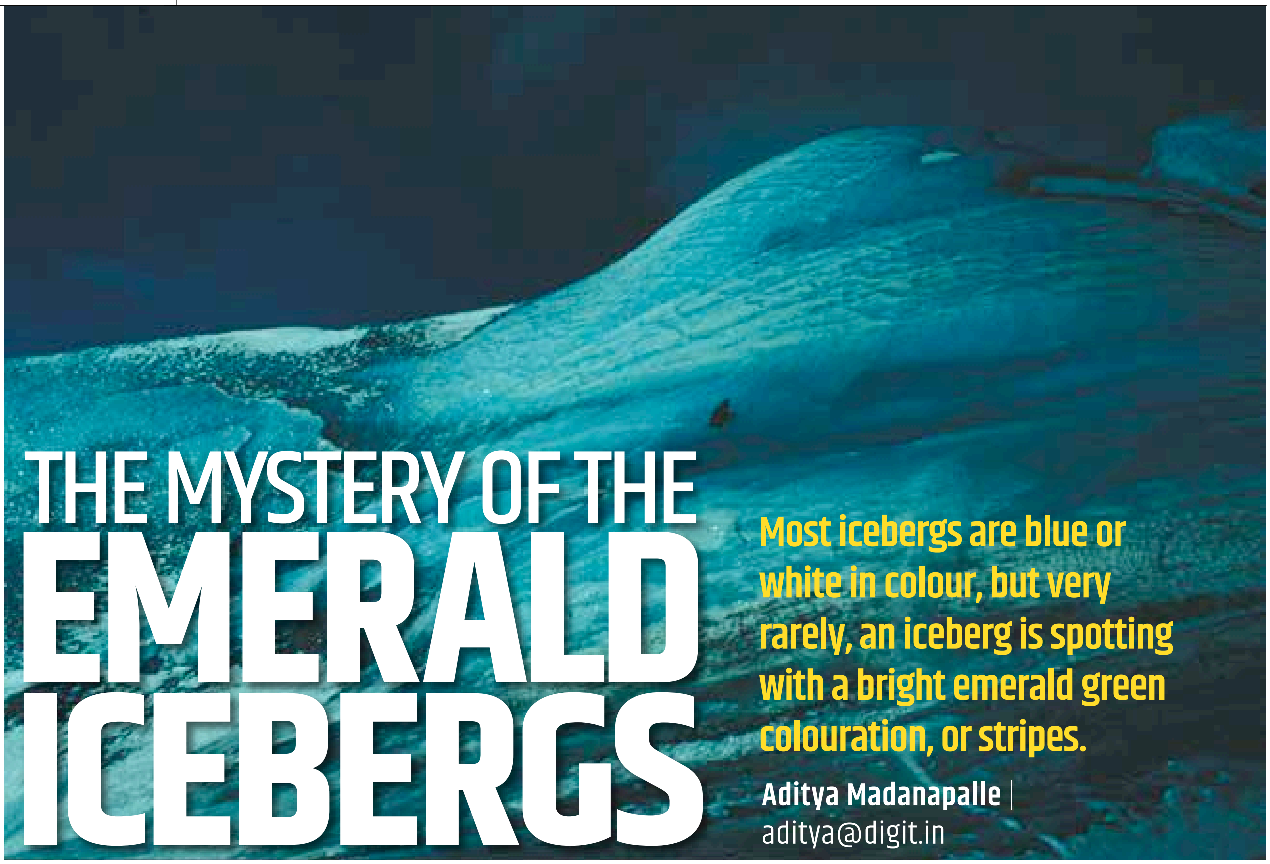


Microbes regulate carbon
Important role by microbes deep within Earth in sequestering carbon over geological timescales. **https://dgit.in/may19-53**



Interstellar meteor strike
'Oumuamua might have zipped past, but it might not have been our first interstellar visitor after all. **https://dgit.in/may19-54**



THE MYSTERY OF THE EMERALD ICEBERGS

Most icebergs are blue or white in colour, but very rarely, an iceberg is spotting with a bright emerald green colouration, or stripes.

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Credit: Collin Roesler.

Icebergs are broken off pieces of glaciers or ice shelves, that float in lakes or seas because they have a lower density than the water. Typically icebergs are white or blue in colouration. If there are a lot of air bubbles in the ice, then all the incident light is scattered equally, giving rise to a white colouration. These kinds of icebergs are formed by layers of snow that solidify over time, and is called glacier ice. Another form of ice, is formed when the sea water in contact with the ice shelves over them. Contact with the floating ice shelves causes the sea water underneath to freeze as well. This kind of ice is compressed with the air bubbles squeezed out, and is called marine ice. As a result, the ice scatters blue light more, while absorbing the red light, giving the iceberg a bluish appearance.

THE UNEXPLAINED COLOURATION

Since the early 1900s, sailors and explorers in Antarctica have reported seeing icebergs with an unexpected colouration – emerald or aqua green. In Australia, they are called Jade Icebergs. The exact process by which these icebergs got their strange colouration, has been a long standing mystery. At times, regular blue or white icebergs can have green striations running along them. One of the best locations to spot the emerald icebergs was off the Amery Ice Shelf in Antarctica.

Investigating icebergs involves drilling into them and obtaining cylindrical portions of ice called "core samples". The properties of these samples are then studied. Studies based on drilled samples showed that the green icebergs are made up of marine ice, with a dark, deep colouration and hardly any air bubbles. It was clear that the emerald icebergs were made up of marine ice. Initially, it was believed that the icebergs had a green colouration because of algal colonies, that grew in the underside of icebergs. Due to the process of the ice melting, the icebergs could topple, exposing the parts that were previously underwater, causing the green coloured parts to be visible. Another theory was that the ice contained dissolved organic carbon, microscopic remains of dead plants and animals. The dissolved organic carbon is yellow in colour, and researchers initially speculated that the yellow combined with the blue of the ice gave rise to the green colouration. However, drilled samples showed that blue icebergs had the same amount of dissolved organic carbon as green icebergs, which meant that something else was giving rise to the colouration. Another proposed theory was that dead phytoplankton was responsible for the green colouration. Again, research ruled out the possibility, as they were not present in significant enough quantities to cause the colouration. Essentially, while the existence of green icebergs were known, scientists had no clue why they had that colour.

Now, new research indicates that the decades old mystery might finally be close to getting solved. The study